

Instagram User Behavior & Engagement Quality Analysis

Product Analytics Dashboard — Portfolio Project

Dataset: 1,547,896 Users | 58 Features | Tools: PostgreSQL · Tableau · Python

1. Executive Summary

Business Challenge

Instagram is shifting its product strategy from maximizing user screen time to improving Engagement Quality Efficiency (EQE) by encouraging meaningful user interactions over passive content consumption. Understanding the behavioral patterns that drive healthy engagement is essential for improving long-term user retention, reducing passive scrolling behavior, and supporting sustainable platform growth.

Key Findings

- Light Users achieved the highest Engagement Quality Efficiency (EQE = 2.74) compared to Heavy and Power Users.
- Casual platform usage recorded the highest EQE (6.21), while Excessive Usage declined to 0.92.
- Quick Feed Users (6.14) and Quick Reel Viewers (4.71) consistently outperformed immersive content consumers.
- Low Stress users achieved significantly higher EQE (2.78) than Very High Stress users (0.87).
- 109,371 users (7.07%) were identified as Dormancy Risk Users, representing an opportunity for targeted retention initiatives.
- Behavioral patterns demonstrated greater variation in EQE than demographic characteristics, indicating that user behavior is a stronger driver of engagement quality within this analysis.

Business Recommendation

Prioritize meaningful engagement over increased screen time by optimizing Feed ranking, encouraging intentional user interactions, reducing passive scrolling behavior, and implementing targeted retention strategies for Dormancy Risk Users. Continuously monitor Engagement Quality Efficiency alongside supporting platform health metrics to enable data-driven product decisions that improve long-term platform health.

2. Business Problem

Situation

Instagram is shifting its product strategy from maximizing user screen time to improving Engagement Quality Efficiency (EQE) by encouraging meaningful user interactions over passive content consumption. As user engagement evolves across Feed, Reels, Explore, Stories, and Direct Messages, understanding the behaviors that drive healthy engagement has become essential for long-term platform growth.

Problem

The platform needs to identify:

- Behavioral patterns associated with high and low Engagement Quality Efficiency.
- User segments exhibiting passive scrolling and excessive platform usage.
- Lifestyle and engagement factors influencing meaningful interactions.
- Users at risk of becoming dormant based on their platform activity.

Business Impact

Without understanding these engagement patterns, Instagram may experience:

- Reduced long-term user retention.
- Increased passive content consumption.
- Lower creator engagement and interaction quality.
- Reduced advertisement effectiveness.
- Missed opportunities for proactive user retention.

Goal

Build an interactive Product Analytics Dashboard that identifies the key drivers of Engagement Quality Efficiency, detects Dormancy Risk Users, and supports data-driven product decisions to improve meaningful engagement, optimize user experience, and strengthen long-term platform health.

3. Business Objectives

The objective of this analysis is to:

- Measure overall platform engagement using Engagement Quality Efficiency (EQE) as the primary success metric.
- Identify the user behaviors that drive high and low engagement quality across the platform.
- Analyze the impact of platform usage, content consumption, and lifestyle factors on meaningful user engagement.
- Detect user segments exhibiting passive consumption and excessive platform usage.
- Identify Dormancy Risk Users to support proactive retention strategies.
- Evaluate overall platform health using supporting engagement metrics.
- Provide actionable insights that support product optimization, personalized user experiences, and long-term platform growth.

4. Decision Makers & Business Questions

Product Manager

Responsibilities

- Improve user engagement quality
- Optimize product features and user experience
- Increase long-term user retention
- Drive data-driven product decisions

Key Questions

- Which user behaviors drive higher Engagement Quality Efficiency (EQE)?
- Which platform experiences encourage passive content consumption?
- What product improvements can increase meaningful user engagement?

Growth & Retention Team

Responsibilities

- Improve user retention
- Reduce user dormancy and churn
- Personalize engagement strategies
- Increase long-term platform activity

Key Questions

- Which users are at risk of becoming dormant?
- Which user segments should be prioritized for retention campaigns?
- How can personalized interventions improve long-term user engagement?

User Engagement & Recommendation Team

Responsibilities

- Optimize Feed and Reels experience
- Improve content recommendations
- Encourage meaningful user interactions
- Reduce passive scrolling behavior

Key Questions

- Which content consumption patterns generate the highest Engagement Quality Efficiency?
- How do Feed and Reels usage influence meaningful engagement?
- Which behavioral patterns should be encouraged to improve overall platform health?

5. Dataset Overview

The analysis was performed using:

- 1,547,896 Instagram Users
- 58 User-Level Features
- Behavioral & Engagement Metrics
- Lifestyle & Demographic Attributes
- Platform Usage Data (Feed, Reels, Explore & Messages)
- Product Analytics & Retention Metrics

6. KPI Framework

Engagement Quality Efficiency (EQE)

Measures the efficiency of converting user attention into meaningful engagement.

Formula:

Meaningful Engagement (ME) ÷ Attention (A)

Purpose:

Evaluate the overall quality of user engagement across the platform.

Comment-to-Like Ratio

Measures the proportion of comments generated relative to likes.

Purpose:

Evaluate the quality of user conversations and interaction depth.

DM Activity Index

Measures the level of private user interactions through Direct Messages.

Purpose:

Evaluate meaningful social connections and private engagement.

Advertisement Click-Through Rate (CTR) Proxy

Measures the proportion of advertisements clicked relative to advertisements viewed.

Purpose:

Evaluate advertisement engagement effectiveness.

Advertisement Load Proxy

Measures advertisement exposure relative to overall user attention.

Purpose:

Evaluate the balance between monetization and user experience.

Dormancy Risk Rate

Measures the percentage of users identified as Dormancy Risk Users based on predefined behavioral conditions.

Purpose:

Evaluate user retention risk and support proactive re-engagement strategies.

7. Analysis Approach

The analysis followed a structured product analytics framework.

Step 1: Evaluate Overall Engagement Quality

- Calculate Engagement Quality Efficiency (EQE)

Step 2: Identify Behavioral Drivers

- Session Frequency Analysis
- Session Duration Analysis
- Daily Platform Usage Analysis
- Feed Consumption Analysis
- Reels Consumption Analysis

Step 3: Analyze User Engagement Patterns

- Posting Behavior Analysis
- Commenting Behavior Analysis
- Direct Messaging Analysis
- Lifestyle & Demographic Analysis

Step 4: Identify High-Value & At-Risk User Segments

- User Behavior Combination Analysis
- Dormancy Risk Identification
- Business Recommendations

8. Dashboard Overview

Engagement Quality Dashboard Provides:

- Overall Engagement Quality Efficiency (EQE)
- Total User Base
- Session Frequency Analysis
- Feed Consumption Analysis
- Reels Consumption Analysis
- Stress Level Analysis

Purpose:

Provide an executive view of the key behavioral factors influencing Engagement Quality Efficiency across the platform.

User Behavior & Retention Dashboard Provides:

- Session Frequency & Feed Consumption Analysis
- Reels Consumption & Stress Analysis
- Platform Usage Intensity Analysis
- Dormancy Risk Distribution
- Supporting Platform Health Metrics

Purpose:

Identify high-value user behaviors, detect Dormancy Risk Users, and support data-driven product optimization and retention strategies.

9. Detailed Findings

Finding 1: Light Session Frequency Is Associated With Higher Engagement Quality

Evidence

Engagement Quality Efficiency (EQE) by Session Frequency:

- Light Users (1–5 sessions/day) = 2.74
- Moderate Users (6–15 sessions/day) = 0.94
- Heavy Users (16–30 sessions/day) = 0.88
- Power Users (31+ sessions/day) = 0.85

Insight

Users with lower daily session frequency achieved substantially higher Engagement Quality Efficiency than users visiting Instagram more frequently.

Business Impact

Increasing visit frequency alone does not improve engagement quality and may encourage lower-value interactions over time.

Recommendation

Prioritize product experiences that encourage meaningful engagement during each session rather than maximizing session frequency.

Finding 2: Shorter Feed Consumption Produces Higher Engagement Quality

Evidence

EQE by Feed Consumption Level:

- Quick Feed Users (2–15 min) = 6.14
- Balanced Feed Users (16–30 min) = 1.57
- Feed Focused Users (31–60 min) = 1.16
- Feed Immersed Users (60+ min) = 0.91

Insight

Users spending less time continuously browsing the Feed consistently achieved higher Engagement Quality Efficiency than users with prolonged Feed consumption.

Business Impact

Extended Feed browsing appears to be associated with passive consumption rather than meaningful engagement.

Recommendation

Optimize Feed ranking to surface relevant content earlier and encourage high-quality interactions within shorter browsing sessions.

Finding 3: Shorter Reel Consumption Is Associated With Better Engagement

Evidence

EQE by Reel Consumption Level:

- Quick Reel Viewers (1–15 min) = 4.71
- Regular Reel Viewers (16–30 min) = 1.23

- Reel Enthusiasts (31–60 min) = 1.00
- Reel Immersed Viewers (60+ min) = 0.87

Insight

Users consuming Reels in shorter viewing sessions demonstrated considerably higher engagement quality than users with prolonged Reel consumption.

Business Impact

Extended Reel viewing may increase passive content consumption, reducing overall engagement efficiency.

Recommendation

Improve Reel recommendations to encourage meaningful interactions instead of prolonged passive viewing.

Finding 4: Lower Stress Levels Are Associated With Higher Engagement Quality

Evidence

EQE by Stress Level:

- Low Stress (Score 0–10) = 2.78
- Moderate Stress (Score 11–20) = 1.18
- High Stress (Score 21–30) = 0.93
- Very High Stress (Score 31+) = 0.87

Insight

Users with lower reported stress levels consistently demonstrated higher Engagement Quality Efficiency than users experiencing higher stress.

Business Impact

Healthier engagement patterns appear to be associated with better user well-being, supporting long-term platform sustainability.

Recommendation

Continue improving personalized recommendations and healthy engagement experiences that encourage positive platform usage.

Finding 5: Casual Platform Usage Delivers The Highest Engagement Quality

Evidence

EQE by Daily Platform Usage:

- Casual Usage (5–30 min/day) = 6.21
- Standard Usage (31–60 min/day) = 1.60
- High Usage (61–120 min/day) = 1.18
- Excessive Usage (120+ min/day) = 0.92

Insight

Users with lower overall daily Instagram usage consistently achieved higher Engagement Quality Efficiency than users with prolonged platform usage.

Business Impact

Simply increasing user screen time is unlikely to improve engagement quality and may encourage passive platform behavior.

Recommendation

Focus product optimization on improving interaction quality rather than increasing total time spent on the platform.

Finding 6: Combined User Behaviors Reveal High-Value Engagement Segments

Evidence

The combination analysis identified the highest Engagement Quality Efficiency among users exhibiting multiple healthy behavioral patterns:

- Light Users + Quick Feed Users = 6.14
- Light Users + Casual Usage = 6.21
- Quick Reel Viewers + Low Stress = 5.00

Conversely, combinations involving Heavy Users, Power Users, Feed Immersed Users, Reel Immersed Viewers, and Very High Stress users consistently recorded lower EQE values.

Insight

Engagement Quality is influenced by combinations of user behaviors rather than any single behavioral metric.

Business Impact

Behavior-based segmentation provides stronger opportunities for personalization and product optimization than analyzing individual metrics independently.

Recommendation

Adopt behavioral segmentation models for content recommendations, Feed ranking, and engagement strategies to improve overall platform performance.

Finding 7: Dormancy Risk Users Represent A Key Retention Opportunity

Evidence

Dormancy Risk Analysis identified:

- Dormancy Risk Users = 109,371 (7.07%)
- Active Users = 1,438,525 (92.93%)

Dormancy Risk Users were identified using low session frequency, low notification responsiveness, and prolonged inactivity (last login > 30 days).

Insight

Although most users remain active, a measurable portion of the user base already exhibits early signs of disengagement.

Business Impact

Without proactive intervention, these users are more likely to reduce platform activity and contribute to future retention challenges.

Recommendation

Implement personalized re-engagement campaigns using targeted notifications and relevant content recommendations for Dormancy Risk Users.

Finding 8: Supporting Platform Health Metrics Complement EQE Monitoring

Evidence

Supporting Platform Health Metrics:

- Comment-to-Like Ratio = 0.29
- DM Activity Index = 8.49
- Ad CTR Proxy = 0.25
- Ad Load Proxy = 0.15

Insight

While EQE serves as the primary engagement metric, supporting metrics provide additional visibility into conversation quality, private interactions, advertising engagement, and advertising exposure.

Business Impact

Monitoring these metrics together enables balanced decision-making across user engagement, monetization, and long-term platform health.

Recommendation

Use EQE alongside supporting platform health metrics as part of a comprehensive product performance monitoring framework.

10. Action Plan

Priority	Action	Owner
High	Optimize Feed ranking to encourage meaningful engagement over passive scrolling.	Product Manager
High	Improve Reel recommendations to increase high-quality user interactions.	Product Manager
High	Launch targeted re-engagement campaigns for Dormancy Risk Users (109,371 users / 7.07%) through personalized notifications and content.	Growth Team
High	Implement behavioral user segmentation to personalize the user experience and improve engagement quality.	Product Analytics Team
Medium	Monitor Engagement Quality Efficiency (EQE = 1.47 overall) as the primary product performance metric across user segments.	Product Analytics Team
Medium	Track Supporting Platform Health Metrics (Comment-to-Like Ratio: 0.29, DM Activity Index: 8.49, Ad CTR Proxy: 0.25, Ad Load Proxy: 0.15) alongside EQE.	Product Analytics Team
Medium	Continuously evaluate user behavior trends and optimize product features to improve long-term engagement quality.	Product Manager

11. Expected Business Outcomes

If the recommendations are implemented:

- Improve Engagement Quality Efficiency (EQE) across key user segments.
- Increase meaningful user interactions through improved Feed and Reel recommendations.
- Reduce Dormancy Risk by proactively re-engaging the 109,371 at-risk users (7.07% of user base).
- Improve long-term user retention through personalized engagement strategies.
- Support healthier platform usage by reducing passive content consumption.
- Enable data-driven product decisions using EQE and Supporting Platform Health Metrics.
- Balance user engagement and monetization while maintaining a positive user experience.
- Strengthen overall platform health through continuous behavioral analytics and performance monitoring.